A simplified unified propagator model for signed order flow, concave price impact and rough volatility

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Abstract

In this note, we simplify the setup in [MORS26] by modelling persistent signed order flow F_t as the difference of two integrated hyper-rough Heston-type processes with an L^1-kernel and Hölder exponent in $(\frac{1}{2},1)$, which endogenously determines a (pure feedback) martingale price process P, the price impact kernel for the propagator equation, and the celebrated square-root impact law in the $H\to 0$ limit. This obviates the need to use nearly unstable heavy-tailed Hawkes processes, and lengthy macroscopic scaling limit arguments with C-tightness as in [MORS26]. We also discuss longer-memory/multiscale behaviour for F, and the addition of proportional bid-ask spreads and exogenous choice of price impact kernel, wherein P is no longer a martingale but the model is still arbitrage-free if P satisfies the conditional full support property in Guasoni et al.[GRS08]. We also propose alternate models where signed order flow is a Gaussian Volterra process with $H\in(\frac{1}{2},1)$, or using fGN which leads to power-law decay for the a.c.f. of the order flow sign process as in [LMF05],[MB25].

1.1 Persistent order flow

Following [AJ21] (and similar to Theorem 3.1 in [MORS26]), we model the cumulative positive order flow F_t^+ for an asset (i.e. the total number of shares bought by time t) as a weak solution to the Stochastic Volterra Equation:

$$F_t^+ \;=\; g(t)+\int_0^t f(t-s)M_s ds \tag{1}$$

where M is a continuous local martingale with $M_0=0$ and $[M]_t=F_t^+$ for $t\geq 0$, $f(t)=\nu e^{-\theta t}t^{H-\frac{1}{2}}$ with $H\in(-\frac{1}{4},0)$, $\nu,\theta>0$ so $f\in L^1$, and $g(t)=V_0 t$ for some $V_0>0$ (note θ replaces the ergodicity parameter λ in Theorem 3.1 in [MORS26]). Weak existence and uniqueness for (1) is established in Theorem 2.2 of [AJ21] in terms of a Volterra Integral Equation (VIE), and F^+ is a non-decreasing, nonnegative, continuous and adapted process. F_t^+ is also a.s. $(2\alpha-\varepsilon)\wedge 1$-Hölder continuous where $\alpha=H+\frac{1}{2}$ (see e.g. Theorem 3.1 in [JR20]), so from the condition on H we see that $2\alpha\in(\frac{1}{2},1)$ [1], and F_t^+ is non-differentiable if $H\leq 0$. Note that if $H>0$, F^+ is just the integrated variance $\int_0^t V_s ds$ for a standard (i.e. non-hyper) Rough-Heston type variance process V, see e.g. section 5.4 of [FGS21] for details.

Lemma 1.1 $\mathbb{E}(F_t^+)<\infty$.

Proof. See Appendix A. ■

Corollary 1.2 *Assume (F^+,M) is a weak solution of (1) in the sense of [AJ21], with $b=0$, $c=1$ and $\nu\equiv 0$. Then M is a continuous square-integrable martingale.*

Proof. See Appendix B. ■

1.2 The inversion formula

If h is a function which solves $h*f\equiv 1$, then we have the inversion formula:

$$\begin{aligned}\int_0^t h(t-s)(F_s^+-g(s))ds \;&=\; \int_0^t h(t-s)\int_0^s f(s-u)M_u du ds \;=\; \int_0^t\int_u^t h(t-s)f(s-u)ds\, M_u du\\ &=\; \int_0^t\int_0^{t-u} h(t-s-u)f(s)ds\, M_u du\\ &=\; \int_0^t (h*f)(t-u)M_u du \;=\; \int_0^t M_u du\end{aligned}$$

(where Fubini is justified since $F_t^+<\infty$ a.s. for finite t and M is continuous), and hence

$$M_t \;=\; \frac{d}{dt}\int_0^t h(t-s)(F_s^+-g(s))ds \;=\; \frac{d}{dt}\int_0^t h(t-s)F_s^+ ds-\frac{d}{dt}\int_0^t h(t-s)g(s)ds \tag{2}$$

[1] we can simulate F^+ using the Monte Carlo scheme in [AA25] using Normal Inverse Gaussian variates

Lebesgue a.e., where the final equality holds since $\int_0^t h(t-s)(F_s^+ - g(s))ds$ and $\int_0^t h(t-s)g(s)ds$ are both absolutely continuous, and hence so is their sum. Moreover, setting $\bar{H}(t) = \int_0^t h(s)ds$, integrating by parts we see that

$$\int_0^t \bar{H}(t-s)dF_s^+ \quad = \quad F_s^+ \bar{H}(t-s)_{s=0}^{s=t} + \int_0^t h(t-s)F_s^+ ds \quad = \quad \int_0^t h(t-s)F_s^+ ds\,.$$

Taking Laplace transforms, the condition $h * f \equiv 1$ becomes $\hat{f}(\lambda)\hat{h}(\lambda) = \frac{1}{\lambda}$, from which we find that

$$h(t) \quad = \quad \frac{\theta^\alpha}{\nu\Gamma(\alpha)}(1 - \frac{\Gamma(-\alpha, t\theta)}{\Gamma(-\alpha)})$$

where $\Gamma(a,z) = \int_z^\infty s^{a-1}e^{-s}ds$ is the incomplete Gamma function, and $h(t) = O(t^{-H-\frac{1}{2}})$ as $t \to 0$. Moreover (from the Final value theorem for Laplace transforms), we see that

$$h(\infty) \quad = \quad \lim_{t\to\infty} h(t) \quad = \quad \lim_{\lambda\to 0} \lambda\hat{h}(\lambda) \quad = \quad \frac{1}{\hat{f}(0)} \quad = \quad \frac{1}{\|f\|_{L^1}} \quad = \quad \frac{\theta^\alpha}{\nu\Gamma(\alpha)}\,. \tag{3}$$

1.3 Signed order flow

Again following [MORS26], we now assume the *signed* order flow for an asset is

$$F_t \quad = \quad F_t^+ - F_t^-$$

where $F_t^- = g(t) + \int_0^t f(t-s)M_s^- ds$ and (M^-, F^-) is an independent copy of (M, F^+). Then we see that

$$\frac{d}{dt}\int_0^t h(t-s)F_s ds \quad = \quad \frac{d}{dt}\int_0^t \bar{H}(t-s)dF_s \quad = \quad \int_0^t h(t-s)dF_s \quad = \quad M_t - M_t^- \tag{4}$$

since the g terms cancel (and the second equality follows from the Stieltjes-Leibniz rule using the known Hölder continuity of $F^\pm$ and that $|h(u)| \le Cu^{-\alpha}$ so $\int_0^t |h(t-s)|(dF_s^+ + dF_s^-) < \infty$), and for $0 \le t \le u$ we have

$$\mathbb{E}(F_u^+|\mathcal{F}_t^M) \quad = \quad g(u) + \mathbb{E}(\int_0^u f(u-s)M_s ds\,|\mathcal{F}_t^M) \quad = \quad g(u) + \int_0^t f(u-s)M_s ds + M_t\int_t^u f(u-s)ds$$

(and similarly for $\mathbb{E}(F_u^-|\mathcal{F}_t^{M^-})$), since $(M_t)_{t\ge 0}$ is an $(\mathcal{F}_t^M)_{t\ge 0}$-martingale by Corollary 1.2, so

$$\mathbb{E}(F_u^+ - F_u^-|\mathcal{G}_t) \quad = \quad \int_0^t f(u-s)(M_s - M_s^-)ds + (M_t - M_t^-)\int_t^u f(u-s)ds \tag{5}$$

where $\mathcal{G}_t = \sigma(\mathcal{F}_t^M, \mathcal{F}_t^{M^-})$.

1.4 Endogenous price dynamics and the propagator equation

We now make the common assumption (cf.[JR20]) that the asset price $P_t = \kappa \lim_{u\to\infty} \mathbb{E}(F_u|\mathcal{G}_t)$ where $F_t = F_t^+ - F_t^-$ (for some $\kappa > 0$) [2]. Then letting $u \to \infty$ in (5), we find that $f(u-s) \searrow 0$ as $u \to \infty$ since $\theta > 0$ so the first term in (5) goes to zero since $|\int_0^t f(u-s)(M_s - M_s^-)ds| \le \sup_{s\in[0,t]} |M_s - M_s^-| \int_0^t f(u-s)ds$ using the monotone convergence theorem, but $\lim_{u\to\infty}\int_t^u f(u-s)ds = \lim_{u\to\infty}\int_0^{u-t} f(r)dr = \int_0^\infty f(r)dr = \|f\|_{L^1} = 1/h(\infty)$ (from (3)), so

$$P_t \quad = \quad \kappa \lim_{u\to\infty} \mathbb{E}(F_u^+ - F_u^-\,|\mathcal{G}_t) \quad = \quad \frac{\kappa}{h(\infty)}(M_t - M_t^-)$$

i.e. P is the difference of two i.i.d. driftless hyper-rough-type Heston models (which effectively both have correlation $\rho = 1$ since M and M^- only have one driving Brownian motion each so P is a "pure feedback" price process, i.e. only depends on its own history), and if $H = 0$ the price process is self-similar with index $\frac{1}{2}$ (see Lemma 3.1 in [FGS21]). We also recover the propagator equation:

$$P_t \quad = \quad \int_0^t G(t-s)dF_s$$

where the price impact kernel G is given by

$$G(t) \quad = \quad \kappa\frac{h(t)}{h(\infty)} \tag{6}$$

and we define the *unsigned* order flow as $V_t = F_t^+ + F_t^-$.

[2] note we are assuming $P_0 = 0$ without loss of generality to ease notation, but we can easily add a positive P_0 term

1.5 Power-law price impact

In particular, the *market impact function* of an exogenous metaorder executed at constant trading speed 1 up to time t_0 is given by

$$MI(t) \quad = \quad \int_0^{t\wedge t_0} G(t-s)ds \tag{7}$$

(where $\alpha_- = \frac{1}{2} - H$) which is $O(t^{\frac{1}{2}-H})$ as $t \to 0$ (for $H \neq \frac{1}{2}$) and globally concave in t for $H \in (-\frac{1}{2}, \frac{1}{2})$, consistent with empirical evidence (note the usual *square-root law* corresponds to $H = 0$ here). Upon cancelling terms, we can show that the asymptotic (i.e. permanent) impact as $t \to \infty$ (given no trading after t_0) is

$$MI(\infty) \quad = \quad \lim_{t\to\infty} \int_0^{t\wedge t_0} G(t-s)\, ds \quad = \quad \int_0^{t_0} (\lim_{t\to\infty} G(t))ds \quad = \quad \kappa t_0 \lim_{t\to\infty} \frac{h(t)}{h(\infty)} \quad = \quad \kappa t_0$$

(this corresponds to the asymptotic grey line in Figure 2 below).

For the special case $H = \frac{1}{2}$ that we have previously excluded, $h(t) = \frac{1}{\nu}(\theta + \delta(t))$, which corresponds to linear permanent price impact plus temporary price impact (coming from the Dirac term); conversely $\lim_{\theta\to 0} h(t) = const. \times t^{-\frac{1}{2}-H}$. Recall from above that $(F_t^+)_{t\geq 0}$ has Hölder regularity $(2\alpha - \varepsilon) \wedge 1$, and empirical evidence in [MORS26] and below suggests that $2\alpha \in (\frac{1}{2}, 1)$ which corresponds to our standing assumption that $H \in (-\frac{1}{4}, 0)$.

1.6 Empirical results

Below we have tabulated estimates of 2α (i.e. the Hölder exponent of $(F_t)_{t\geq 0}$) using signed order flow for major US tech stocks using the model-independent $\hat{H}^{\pi}_{L,K}(X)$ estimator in Eq 7 in Cont&Das[CD24] which is consistent and allows for irregularly spaced data (with their $L/K = [\sqrt{N}]$, where N is the total sample size). For this we have used the free data at `https://data.lobsterdata.com/info/DataSamples.php`:

	GOOG	AAPL	MSFT	AMZN	INTC
N	49284	118326	410881	57360	404496
$2\hat{\alpha} = 2(\hat{H} + \frac{1}{2})$	0.692	0.720	0.847	0.735	0.890
$\hat{H}$	-0.154	-0.140	-.0767	-0.133	-0.0553

Note we only use columns 4 and 5 from the lobsterdata.csv files which correspond to actual realized trades.

Try Han-Schied SSE for non-uniform grid

1.7 Controlling liquidity

In the setup above, we have no control over the depth of liquidity, since the kernel $G(t)$ which controls the amount of transient price impact is endogenously determined via the f function in (1). To resolve this inflexibility, we can make the natural assumption that $P_t = \eta\tilde{M}_t + (1-\eta)\int_0^t G(t-s)dF_s$ for $\eta \in [0,1]$ where $\tilde{M}_t$ is an additional martingale, so η close to 1 implies greater liquidity and vice versa.

1.8 Using a mixed kernel

If we augment f to a function of the form $f(t) = \nu e^{-\theta t} t^{H-\frac{1}{2}} + \nu_2 e^{-\theta_2 t} t^{H_2 - \frac{1}{2}}$, then we find that

$$h(t) \quad = \quad \frac{1}{\nu\Gamma(\alpha)} \sum_{n=0}^{\infty} (-c)^n \sum_{k=0}^{\infty} \frac{(\beta n)_k}{k!}\, \delta^k\, \theta^{-(\mu_n + k)}\, \frac{\gamma(\mu_n + k, \theta t)}{\Gamma(\mu_n + k)}$$

where $\beta = \frac{1}{2} + H_2$, $c = \frac{\nu_2\Gamma(\beta)}{\nu\Gamma(\alpha)}$, $\delta = \theta - \theta_2$, $\mu_n = \beta n - \alpha(n+1)$, $\gamma(a,z) = \int_0^z s^{a-1}e^{-s}ds$ is the analytic continuation of the lower incomplete Gamma function and $(a)_k$ denotes the Pochhammer symbol. We can then use two (possibly different) f functions to drive $F^{\pm}$, and if $H_2 \in (0, \frac{1}{2})$, P_t will have an additional standard (i.e. non-hyper) rough component, which may be more realistic. We can also relax the assumption that M and M^- are independent to give greater control over the skewness of P_t.

1.9 Tempered long memory/multiscale behaviour

If we want $(F_t^{\pm})_{t\geq 0}$ to have longer memory/multiscale behaviour (as reported empirically in [MORS26]), we can replace f with e.g. $f(t) = (\frac{1}{c_1 t^{H-\frac{1}{2}}} + \frac{1}{c_2} t^{\gamma})^{-1}$ with $H \in (-\frac{1}{2}, 0)$, $\gamma > 1$ and $c_1, c_2 > 0$ (note f is still in L^1).

1.10 Power-law decay for the acf of order flow signs

We can also consider a discrete-time formulation where signed order $F_n = \nu B_n^H$ for $n = 1, 2, ...$, where B^H is an fBM with $H \in (\frac{1}{2}, 1)$, and let $Z_n = B_n^H - B_{n-1}^H \sim N(0,1)$ so $(Z_n)_{n=1}^\infty$ is fractional Gaussian noise (fGN) with $Z_n \sim N(0,1)$ for all n, then

$$\begin{aligned} \rho(n) \;:=\; \mathbb{E}(Z_{k+n}Z_k) &= \mathbb{E}((B_{k+n}^H - B_{k+n-1}^H)(B_k^H - B_{k-1}^H)) \\ &= \tfrac{1}{2}((n+1)^{2H} - 2n^{2H} + (n-1)^{2H}) \;\sim\; const. \times n^{2H-2} \qquad (n \to \infty) \end{aligned}$$

and we see that Z is stationary. For two correlated $N(0,1)$ random variables X, Y with $\mathbb{E}(XY) = \rho$ it can be shown that $\mathbb{E}(\mathrm{sgn}(X)\mathrm{sgn}(Y)) = \frac{2}{\pi}\arcsin\rho$. Let $\varepsilon_n = \mathrm{sgn}(Z_n)$. Then

$$C(n) \;=\; \mathbb{E}(\varepsilon_k\varepsilon_{k+n}) \;=\; \frac{2}{\pi}\arcsin\mathbb{E}(Z_kZ_{k+n}) \;=\; \frac{2}{\pi}\arcsin\mathbb{E}(Z_0Z_n)\,.$$

But $\arcsin x = x + O(x^3)$ as $x \to 0$, so

$$\mathbb{E}(\varepsilon_k\varepsilon_{k+n}) \;\sim\; \frac{2}{\pi}\mathbb{E}(Z_kZ_{k+n}) \;\sim\; const. \times n^{2H-2}$$

as $|n| \to \infty$, so we have power-law decay for the a.c.f. of the signs of the order flow increments, similar to the metaorder models in [LMF05] and [MB25].

1.11 Bid-ask spreads, exogenous choice of impact kernel, and non-semimartingale price dynamics

We can also include a proportional transaction cost $\varepsilon P_t > 0$ per share bought (i.e. a bid-ask spread). If we then also abandon the (questionable) assumption that $P_t = \kappa \lim_{u\to\infty}\mathbb{E}(F_u^+ - F_u^- \,|\mathcal{G}_t)$, and instead exogenously choose the price impact kernel G and assume that e.g. $P_t = P_0 + \int_0^{t\wedge\tau_\delta} G(t-s)dF_s$ for some deterministic function G where $\tau_\delta = \inf\{t : P_t = \delta\}$ and $P_0 > \delta > 0$ with $\delta \ll 1$; then if P has the conditional full support property, the model is arbitrage-free even if P is not a semimartingale (see e.g. Guasoni et al.[GRS08] for details). Note also that the trading costs due to G are quadratic.

1.12 Automated Market Makers

Let x and y denote the amounts of two different coins in a constant product market maker liquidity pool. The basic Uniswap v1 protocol imposes that $xy = \kappa^2$ where κ is known as the *depth* of the pool, which is constant unless liquidity is added or removed from the pool by liquidity providers (LPs). If a Liquidity taker (LT) sells Δy coins to the pool, then the amount of asset X in the pool changes by amount $\frac{\kappa^2}{y+\Delta y} - \frac{\kappa^2}{y}$, and dividing this by Δy, and letting $\Delta y \to 0$, we see that the rate for an infinitesimal amount of Y is $z = \frac{\kappa^2}{y^2} = \frac{xy}{y^2} = \frac{x}{y}$. We can then assume that the amount of Y coins in the pool at time t is $Y_t = Y_0 + F_t^+ - F_t^-$ until Y hits zero, so $Z_t = \frac{\kappa^2}{Y_t^2}$. In practice, there are also proportional transaction costs of $\varepsilon = .003$, so as in subsection 1.11, if Z satisfies the conditional full support property, the model is arbitrage-free (note we are not assuming existence of an additional traditional LOB-based exchange like Binance here).

1.13 Signed order flow as a persistent Gaussian process

Let $\bar{F}(t) = \int_0^t f(s)ds$ where $f(.)$ is defined as in subsection 1.1 with $H \in (-\frac{1}{4}, 0)$, and we now assume the signed order flow F follows the Gaussian Volterra process:

$$F_t \;=\; \int_0^t \bar{F}(t-s)dW_s \;=\; \bar{F}(t-s)W_s|_{s=0}^{s=t} + \int_0^t f(t-s)W_s ds \;=\; \int_0^t f(t-s)W_s ds$$

where W is a standard Brownian motion. Then $\bar{F}(t) = O(t^{H+\frac{1}{2}}) = O(t^{H+1-\frac{1}{2}})$ as $t \to 0$, i.e. F has Hölder continuity $H + 1 - \varepsilon \in (\frac{1}{2}, 1)$, and

$$\mathbb{E}(F_u|\mathcal{F}_t^W) \;=\; \int_0^t f(u-s)W_s ds + W_t\int_t^u f(u-s)ds$$

and $\lim_{u\to\infty}\mathbb{E}(F_u|\mathcal{F}_t^W) = c_{H,\nu,\theta}W_t$ and $W_t = \int_0^t h(t-s)dF_s$ (interpreted as a Young integral), so the model is plausible but leads to boring Bachelier dynamics for the price process P_t (it does have the advantage that we can use MLE or GMM methods to estimate H and θ). Or we can replace W with another martingale (e.g. a process with rough volatility).

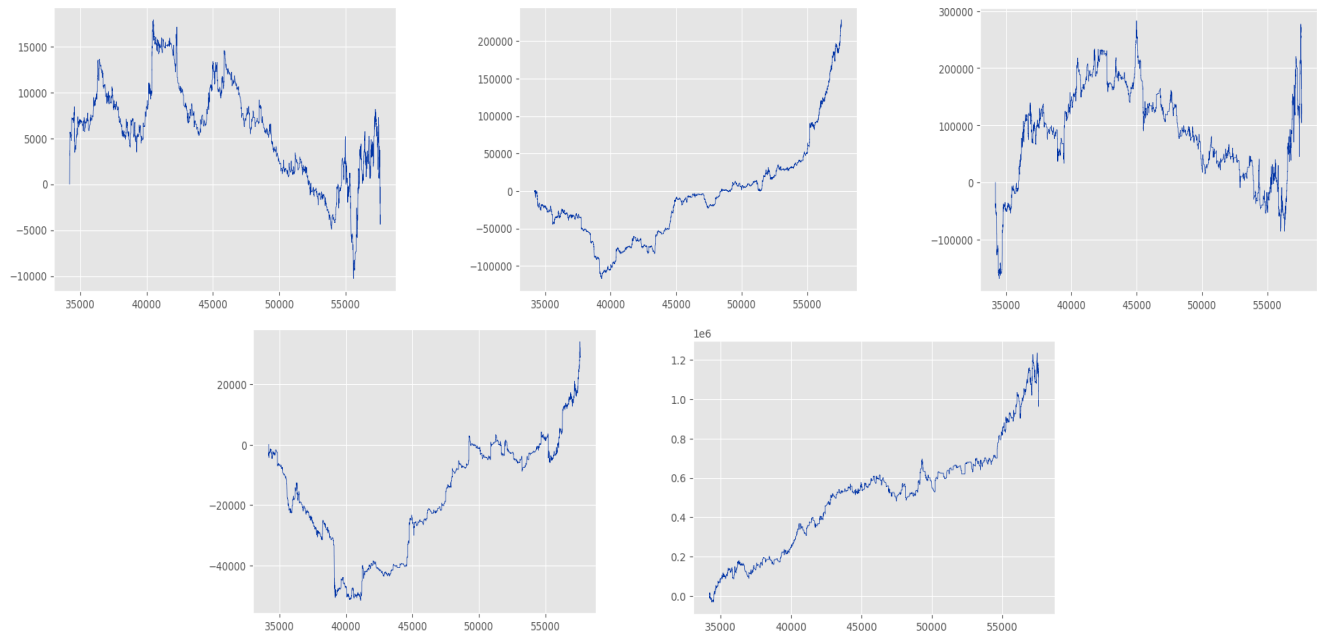

Figure 1: Signed order flow for GOOG, AAPL, MSFT, AMZN and INTC

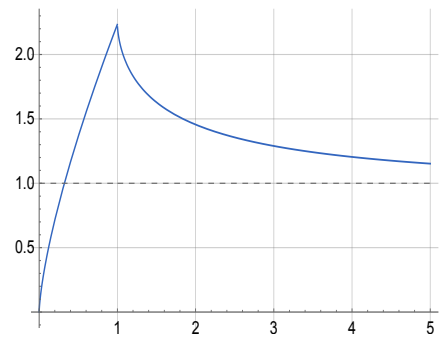

Figure 2: Concave price impact of an exogenous metaorder executed at constant trading speed 1 over $[0, t_0]$ with $H = -0.2$, $\theta = 0.1$, $\kappa = 1$ and $t_0 = 1$. The blue line asymptotes to the constant level $MI(\infty) = \kappa t_0 = 1$ in this case (grey line) as $t \to \infty$ which represents the asymptotic permanent price impact.

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A Appendix A

By Dambis–Dubins–Schwarz, there exists a Brownian motion B (on a possibly enlarged probability space) such that $M_t = B_{F_t^+}$. Now set $G_t := \sup_{0\le r\le t} |g(r)|$, $K_t := \int_0^t |f(r)|dr > 0$, and for $n \ge 1$, define the level hitting time $\tau_n := \inf\{s \ge 0 : F_s^+ \ge n\}$.

Since F^+ is continuous and nondecreasing, on $\{\tau_n \le t\}$ we have $F_{\tau_n}^+ = n$ and $F_r^+ \le n$ for all $0 \le r \le \tau_n$. Fix $n \ge 1$ and work on the event $\{\tau_n \le t\}$. For $0 \le r \le \tau_n$ we have $|B_{F_r^+}| \le \sup_{0\le u\le n} |B_u|$. Setting $s = \tau_n$ and taking absolute values we see that

$$\begin{aligned} n \;=\; g(\tau_n) + \int_0^{\tau_n} f(\tau_n - r)\, B_{F_r^+}\, dr \;&\le\; |g(\tau_n)| + \int_0^{\tau_n} |f(\tau_n - r)|\, |B_{F_r^+}|\, dr \\ &\le\; G_t + (\int_0^{\tau_n} |f(\tau_n - r)|\, dr) \sup_{0\le u\le n} |B_u| \\ &=\; G_t + (\int_0^{\tau_n} |f(s)|\, ds) \sup_{0\le u\le n} |B_u| \;\le\; G_t + K_t \sup_{0\le u\le n} |B_u|. \end{aligned}$$

Hence for $n > G_t$, on $\{\tau_n \le t\}$ we have $\{\sup_{0\le u\le n} |B_u| \ge \frac{n-G_t}{K_t}\}$. Since $F_t^+ \ge n$ iff $\tau_n \le t$, we obtain

$$\mathbb{P}(F_t^+ \ge n) \;\le\; \mathbb{P}(\sup_{0\le u\le n} |B_u| \ge \frac{n - G_t}{K_t}), \qquad n > G_t.$$

By the standard reflection principle bound $\mathbb{P}(\sup_{0\le u\le n} |B_u| \ge a) \le 4\exp(-\frac{a^2}{2n})$, for $a > 0$, so for $n > G_t$,

$$\mathbb{P}(F_t^+ \ge n) \;\le\; 4\exp(-\frac{(n - G_t)^2}{2nK_t^2}).$$

In particular, if $n \ge 4G_t$ then $(n - G_t)^2/n \ge \frac{1}{2}n$, hence

$$\mathbb{P}(F_t^+ \ge n) \;\le\; 4e^{-\frac{n}{4K_t^2}}, \qquad n \ge 4G_t. \tag{A-1}$$

Since $F_t^+ \ge 0$, we also know that $\mathbb{E}(F_t^+) = \int_0^\infty \mathbb{P}(F_t^+ \ge x)\, dx$. Now set $m := \lceil 4G_t \rceil$. Then

$$\mathbb{E}(F_t^+) \;\le\; m + \int_m^\infty \mathbb{P}(F_t^+ \ge x)\, dx \;=\; m + \sum_{n=m}^\infty \int_n^{n+1} \mathbb{P}(F_t^+ \ge x)\, dx \;\le\; m + \sum_{n=m}^\infty \mathbb{P}(F_t^+ \ge n). \tag{A-2}$$

Using (A-1) for all $n \ge m$,

$$\sum_{n=m}^\infty \mathbb{P}(F_t^+ \ge n) \;\le\; \sum_{n=m}^\infty 4\exp(-\frac{n}{4K_t^2}) \;<\; \infty. \tag{A-3}$$

Combining (A-2) and (A-3) yields $\mathbb{E}(F_t^+) < \infty$.

B Appendix B

By the weak-solution formulation, M is a continuous local martingale and $[M]_t = F_t^+$ for $t \ge 0$. Fix $t \ge 0$, and for $n \ge 1$ define

$$\tau_n := \inf\{s \ge 0 : [M]_s \ge n\}.$$

Then $(M_{s\wedge\tau_n})_{s\ge 0}$ is a continuous square-integrable martingale and (in particular) for every $0 \le s \le t$,

$$\mathbb{E}(M_{t\wedge\tau_n} \mid \mathcal{F}_s^M) = M_{s\wedge\tau_n}. \tag{B-1}$$

Moreover, by the Itô isometry for continuous martingales,

$$\mathbb{E}(M_{t\wedge\tau_n}^2) \;=\; \mathbb{E}([M]_{t\wedge\tau_n}) \;\le\; \mathbb{E}([M]_t) \;=\; \mathbb{E}(F_t^+) \;<\; \infty$$

where the last inequality follows from the previous lemma. Hence $(M_{t\wedge\tau_n})_{n\ge 1}$ is uniformly bounded in L^2, and therefore uniformly integrable. Since $[M]_t < \infty$ a.s. for fixed t, we have $\tau_n \to \infty$ a.s., and since M is continuous,

$$M_{t\wedge\tau_n} \to M_t, \qquad M_{s\wedge\tau_n} \to M_s$$

a.s. as $n \to \infty$. By uniform integrability, both convergences also hold in L^1. Passing to the limit in (B-1) yields

$$\mathbb{E}(M_t|\mathcal{F}_s^M) = M_s, \qquad 0 \le s \le t.$$

Thus M is a martingale. Finally, by Fatou's lemma,

$$\mathbb{E}(M_t^2) \;\le\; \liminf_{n\to\infty} \mathbb{E}(M_{t\wedge\tau_n}^2) \;\le\; \mathbb{E}(F_t^+) \;<\; \infty$$

so M is square-integrable.